# Surface ion trap: findings and recommendation

## 151.75 meV predicted RF-only depth

Symmetric 200 µm RF rails • 5 µm gaps
5 µm gold • 500 µm fused silica

100 V peak | 50.141 MHz | q = 0.395

Source: Project finite-gap thickness screen: thickness_screen_results.json

# Objective and constraints

| Constraint | Limit |
| --- | --- |
| Height / radial frequency | 50–100 μm / ≥1 MHz |
| Mathieu / voltage | q ≤0.4 / ≤100 V peak |
| RF width / gap | 20–200 μm / 5–20 μm |
| Drive frequency | 10–60 MHz |

**Extended design: center width and layer thicknesses may vary.**

Source: starter_file_given.ipynb — parameter and score cells

Source: Working notebook — revised center and metal thickness

# Notebook development: three key steps

- Start: lower height and lower RF frequency to increase depth.
- Failure: low-frequency trials exceeded the Mathieu limit.
- Revision: correct the 3.68 ratio and retune frequency near q = 0.4.

Recorded milestone: ~150.5 meV → current saved output: 151.622 meV.

Source: Traped ion working file.ipynb — notes in cells 3–9; saved outputs 20 and 24

# Why the final ratio differs from 3.68

## Depth □ $V^2/f^2$ q □ V G/$f^2$

- 3.68 means RF width / RF separation—not fabrication gap.
- The paper holds height and RF settings fixed; the constrained search varies them.
- Selected effective ratio: (200 + 5)/(26.71361 + 5) ≈ 6.46.

Source: [1] Nizamani & Hensinger, Appl. Phys. B (2012), doi:10.1007/s00340-011-4803-x

Source: Project constrained analytic search: optimize_trap.py

SYMMETRY COMPARISON

# Symmetric rails led the depth search

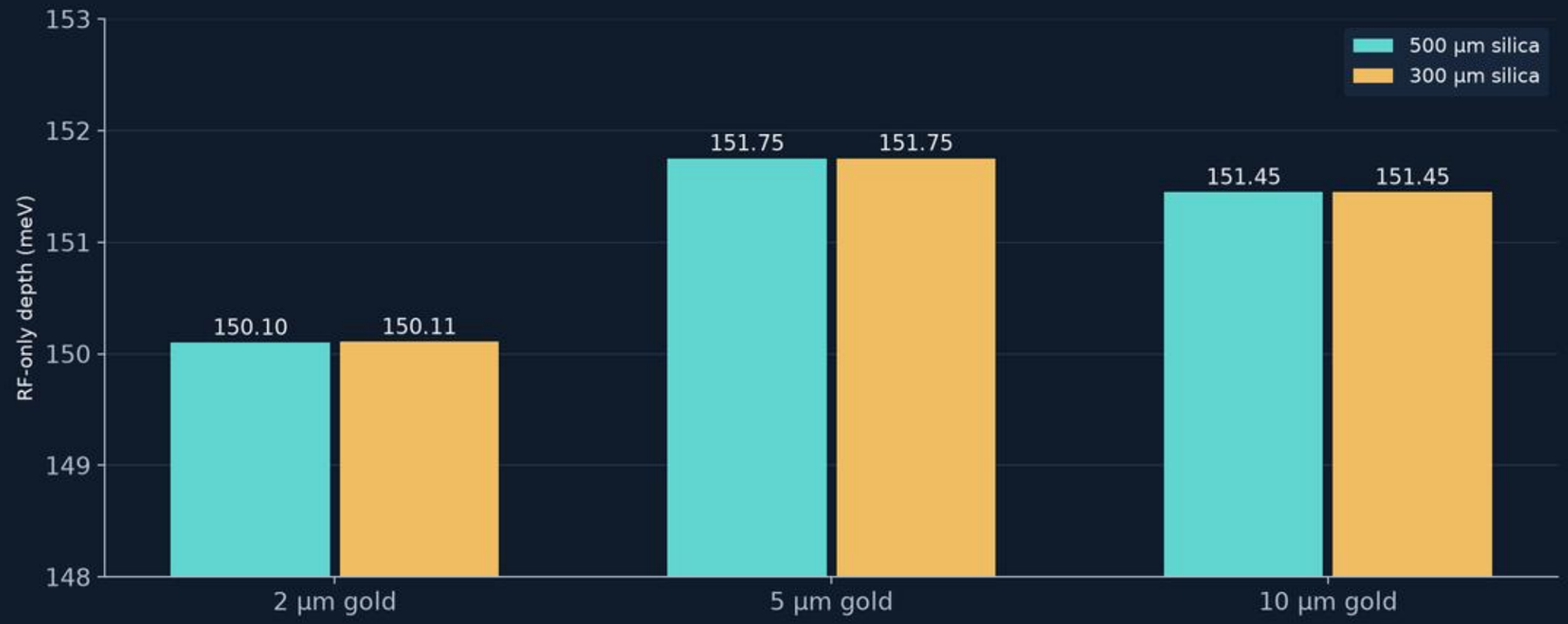
| RF rails; center =60 µm | Predicted depth |
| --- | --- |
| Equal: 200 / 200 µm | 144.29 meV |
| Independent-width search: ≈200 / 200 µm | 144.29 meV |
| Illustrative unequal: 200 / 100 µm | 134.20 meV |

**Asymmetry can aid mode orientation; it did not improve this modeled depth objective.**

Source: Project gapless-plane results: trap_optimization_results.json

Source: [2] Allcock et al., New J. Phys. (2010), doi:10.1088/1367-2630/12/5/053026

# Thickness: retain 5 µm gold on 500 µm silica




Wafer thinning offered negligible benefit; the 5 vs. 10 µm ranking needs a finer mesh.

Source: Project six-case finite-gap field results; retuned to q =0.395

Source: [3] Britton, microfabrication dissertation (2008) — plated-gold fabrication precedent

# Cu/Au may reduce loss—not ideal trap depth

## ~23% lower estimated electrode resistance

- Candidate: 4.8 µm copper + 0.2 µm gold; same 5 µm total thickness.
- At the same 100 V electrode amplitude: no demonstrated depth gain.
- Film quality, coating integrity and RF feed losses require validation.

Source: [6] Tao et al., working Cu/Au surface trap, arXiv:2002.11470 (2020)

Source: [7] Matula, J. Phys. Chem. Ref. Data (1979); resistance estimates in rf_material_comparison.json

# Final candidate and saved notebook result

| Parameter | Recommended screened candidate |
| --- | --- |
| RF width / gap / center | 200 / 5 / 26.71361 μm |
| Gold / silica thickness | 5 / 500 μm |
| Voltage / frequency / q | 100 V / 50.141 MHz / 0.395 |
| Depth / height / radial minimum | 151.75 meV / 58 μm / 6.945 MHz |

Notebook saves 151.622 meV at 49.4 MHz; domain and q differ, so compare cautiously.

Source: Fixed-window field screen + frequency scaling: recommended_thickness_candidate.json

Source: Current saved notebook outputs — inspected, not re-executed for this deck

# What remains before fabrication

- Verify the final drive point with mesh and domain convergence.
- Include axial DC confinement and the full escape barrier.
- Check finished gaps, adhesion, post-bake coating and breakdown.
- Measure the RF network, voltage balance, power loss and temperature.

Source: [5] Schmied, finite-electrode and dielectric corrections, New J. Phys. (2010)

Source: [11] Copper diffusion through gold coatings, Anal. Chim. Acta (2023)